

Algebra 1

Unit 8

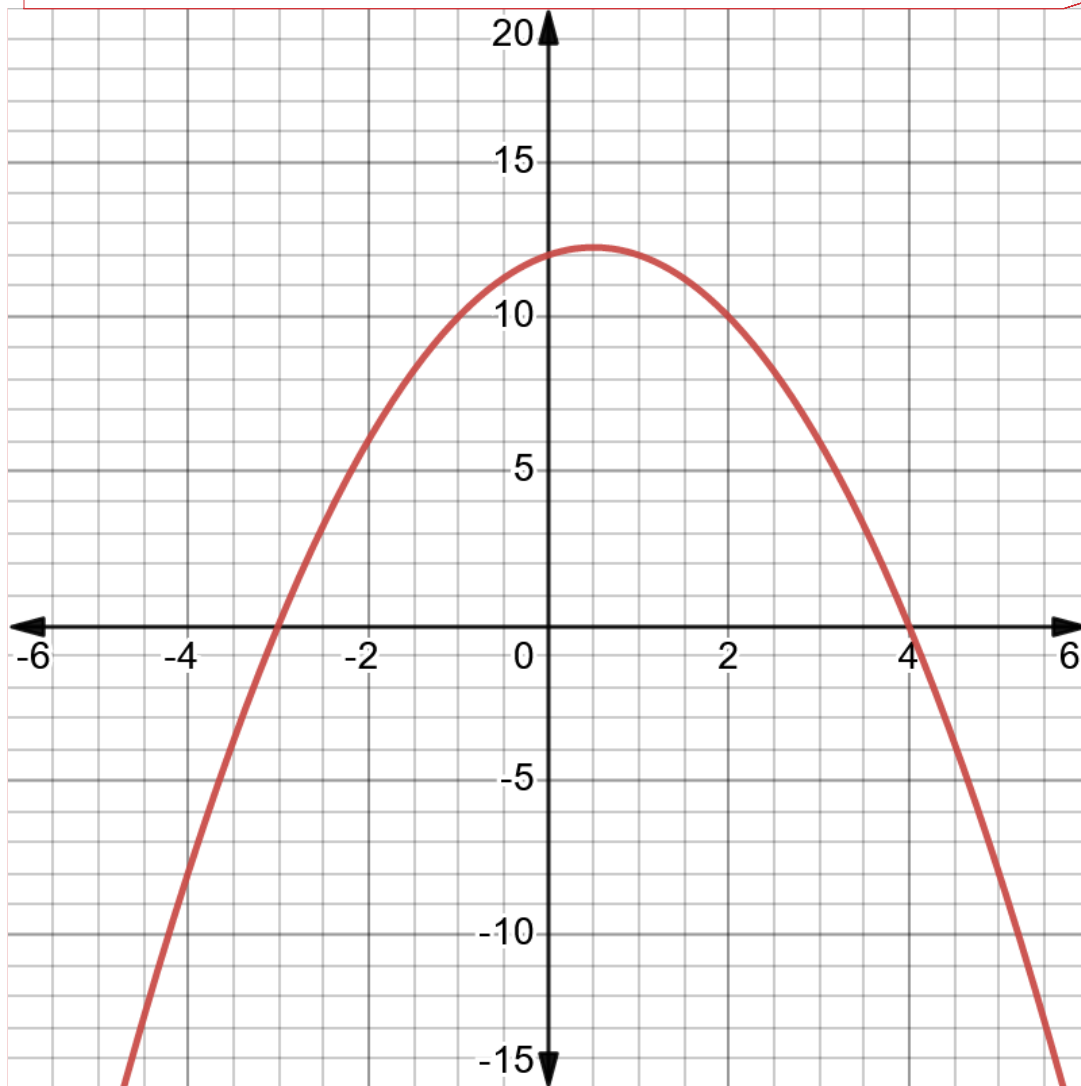
Lesson 4 Homework

Name **Answer Key** _____

Date _____

Period _____

1. Graph the equation $y = -(x - 4)(x + 3)$.



Commented [ND1]: <https://www.desmos.com/calculator/ubei46qytt>

2. Write the key features that can be determined from the equation.

$y = -(x - 4)(x + 3)$	
x-intercept	$(-3,0) (4,0)$
y-intercept	$(0,12)$
Vertex	$(0.5,12.25)$
Domain	All real #'s
Range	$y \leq 12.25$
Left end behavior	$x \rightarrow -\infty \quad y \rightarrow -\infty$
Increasing	$x < 0.5$
Decreasing	$x > 0.5$
Positive	$-3 < x < 4$
Negative	$x < -3, x > 4$
Opens up/down	Down
Axis of symmetry	$x = 0.5$
Right end behavior	$x \rightarrow \infty \quad y \rightarrow -\infty$